EXPLORATORY DATA ANALYSIS REPORT

Titanic Survival Prediction Dataset

Machine Design Document (MDD)

Prepared by: K. Sreejani
Course: Data Science (Data Analytics)
Date: 27-02-2026

1. INTRODUCTION

Source: Titanic Dataset – Kaggle

URL: "titanic-dataset"

Objective:

The objective of this document is to perform an Exploratory Data Analysis (EDA) on the Titanic dataset to understand the factors that influenced passenger survival during the disaster. The dataset used for this analysis is sourced from Kaggle and contains demographic and travel-related information of passengers aboard the Titanic.

Through this analysis, we aim to answer the central question:

"What were the major factors that affected the survival of a passenger on the Titanic?"

The analysis will examine the impact of multiple features such as gender, passenger class, age, fare, and family size. In addition, interaction effects between these variables will also be explored to understand conditional survival patterns.

2. DATASET DESCRIPTION

2.1 Dataset Overview

The dataset used for this analysis is the Titanic passenger dataset, which contains demographic and travel-related information of individuals who were aboard the Titanic.

2.2 Feature Description

Feature	Datatype	Description
1. PassengerId	Numerical (int)	Unique identifier for each passenger
2. Survived	Categorical (binary)	Survival status (0 = Did not survive, 1 = Survived)
3. Pclass	Categorical (ordinal)	Passenger class (1 = First, 2 = Second, 3 = Third)
4. Name	Categorical (object)	Full name of the passenger
5. Sex	Categorical (nominal)	Gender of the passenger
6. Age	Numerical (continuous)	Age of the passenger
7. SibSp	Numerical (continuous)	Number of siblings/spouses aboard
8. Parch	Numerical (continuous)	Number of parents/children aboard
9. Ticket	Categorical (object)	Ticket number
10. Fare	Numerical (float)	Ticket fare paid
11. Cabin	Categorical (object)	Cabin number (if assigned)
12. Embarked	Categorical (object)	Port of embarkation (C, Q, S)

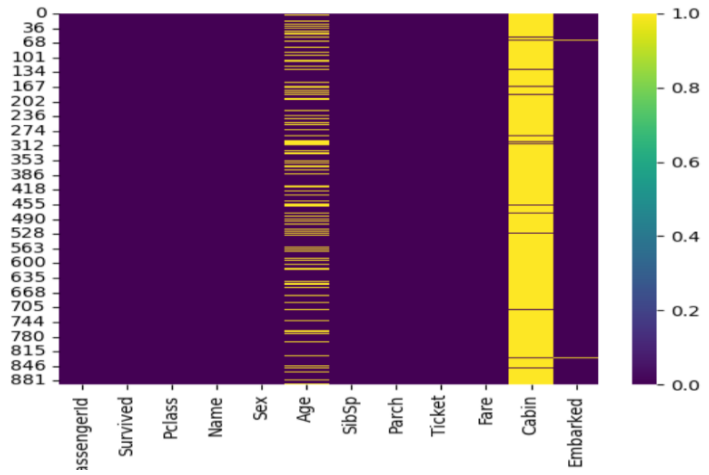
3. DATA CLEANING AND PREPROCESSING

3.1 Handling Missing Values

The initial data exploration revealed missing values in the Age, Cabin, and Embarked features. Since each feature had a different proportion and significance, tailored strategies were applied for handling missing values to preserve data integrity and analytical relevance.

```
[120]: missing_values = data[data.isnull().any(axis = 1)]
       sns.heatmap(missing_values.isnull(), cmap = 'viridis', cbar = True)

[120]: <Axes: >
```



3.1.1 Handling Missing Values in Cabin

The Cabin feature contained a very high proportion (75%) of missing values, making it unreliable for meaningful analysis. Additionally, cabin identifiers are sparse categorical values with insufficient information to meaningfully impute. Therefore, there is a risk of biased inference and the Cabin column was removed from the dataset.

3.1.2 Handling Missing Values in Age

The Age feature had moderate missing values and is an important demographic factor influencing survival. Since the distribution of Age contained outliers, the median value was used for imputation instead of the mean to avoid distortion caused by extreme values.

3.1.3 Handling Missing Values in Embarked

The Embarked feature contained only a few missing values. As it is a categorical variable, the mode (most frequent category) was used for imputation to maintain consistency in categorical distribution.

```
[121]: data.drop('Cabin', axis=1, inplace=True)

[122]: data['Age'].fillna(data['Age'].median(), inplace=True)

[123]: data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

[124]: data.isnull().sum()

[124]: PassengerId    0
       Survived       0
       Pclass         0
       Name           0
       Sex            0
       Age            0
       SibSp          0
       Parch          0
       Ticket         0
       Fare           0
       Embarked       0
       dtype: int64
```

3.2 Feature Engineering

Feature engineering was performed to derive more meaningful variables from existing features. These engineered features help capture hidden relationships and improve interpretability during analysis.

3.2.1 Creating 'Family Size' Feature

The FamilySize feature was created by combining SibSp (siblings/spouses aboard) and Parch (parents/children aboard) along with the passenger themselves. This feature helps analyze whether traveling alone or with family influenced survival chances.

3.2.2 Creating Age Group Feature

The Age feature was transformed into categorical age groups to better understand survival trends across different life stages. This categorization helps compare survival priorities such as children, adults, and senior passengers during evacuation.

```
dtype: int64
```

FEATURE ENGINEERING

```
[14]: data['FamilySize'] = data['SibSp'] + data['Parch'] + 1
```

```
[15]: data['AgeGroup'] = pd.cut(
          data['Age'],
          bins=[0, 12, 20, 40, 60, 80],
          labels=['Child', 'Teen', 'Adult', 'MiddleAge', 'Senior']
      )
```

```
[16]: data.columns
```

```
[16]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
             'Parch', 'Ticket', 'Fare', 'Embarked', 'FamilySize', 'AgeGroup'],
            dtype='object')
```

3.3 Encoding Categorical Variables

Categorical variables were encoded to prepare the dataset for numerical analysis. Different encoding techniques were applied based on the nature of each feature.

- **Encoding Sex (Label Encoding)**
  The Sex feature contains only two categories, making label encoding suitable. The values were mapped as male = 0 and female = 1 for easier numerical comparison during analysis.
- **Encoding Embarked (One-Hot Encoding)**
  The Embarked feature contains multiple categories with no ordinal relationship. Therefore, one-hot encoding was applied to convert each category into separate binary columns, preventing the introduction of artificial ordering.

ENCODING CATEGORICAL VARIABLES

```
[17]: data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

```
[23]: data.Sex.unique()
```

```
[23]: array([0, 1])
```

```
[18]: data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
[24]: data.columns
```

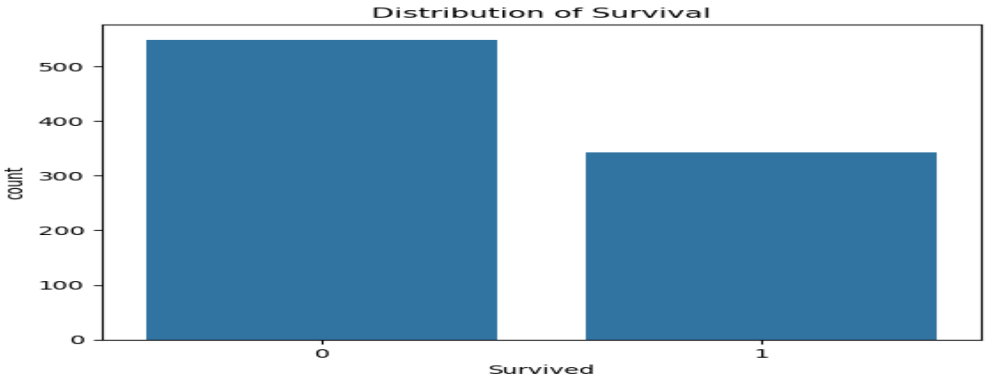
```
[24]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
             'Parch', 'Ticket', 'Fare', 'FamilySize', 'AgeGroup', 'Embarked_Q',
             'Embarked_S'],
            dtype='object')
```

After completing data cleaning and preprocessing, the dataset became suitable for reliable exploratory analysis. The following sections present univariate, bivariate, and multivariate analyses performed on the cleaned dataset to derive meaningful insights about survival patterns.

4. UNIVARIATE ANALYSIS

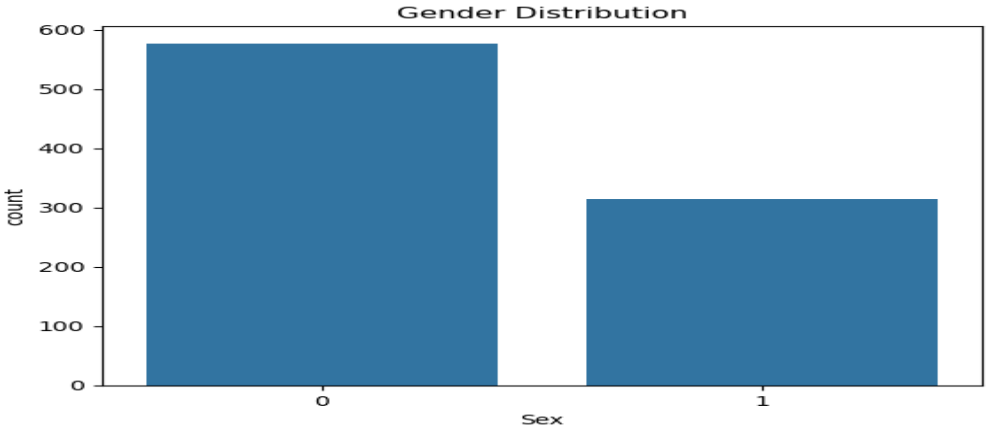
4.1 Survival Distribution

The survival distribution shows that the number of passengers who did not survive is higher than those who survived. This indicates that overall survival chances during the disaster were relatively low. The imbalance in the target variable highlights the severity of the event and sets the context for further feature-based survival analysis.



4.2 Gender Distribution

The gender distribution indicates the proportion of male and female passengers aboard the Titanic. Understanding this distribution is important because gender is expected to significantly influence survival outcomes due to evacuation priorities during the disaster.

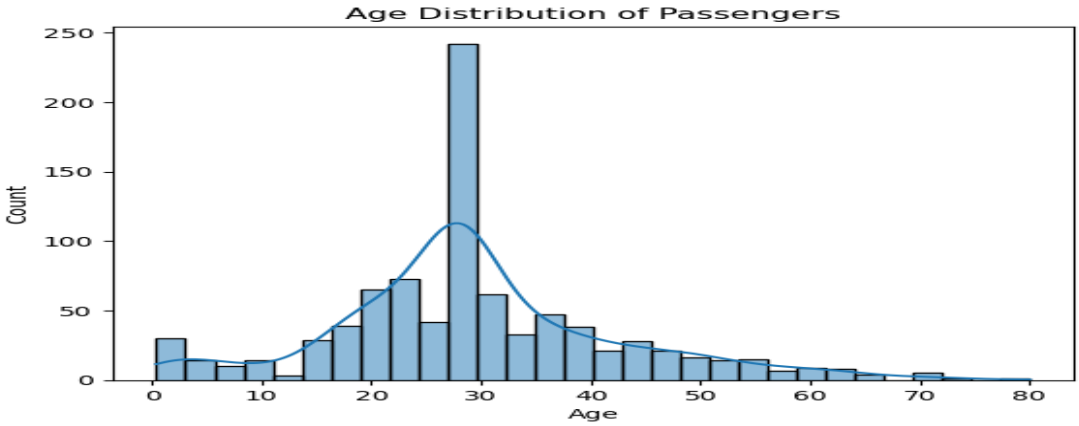


4.3 Passenger Class Distribution

The passenger class distribution shows the number of passengers belonging to first, second, and third classes. This reflects the socio-economic composition of passengers and is expected to influence survival chances due to differences in cabin location and access to lifeboats.

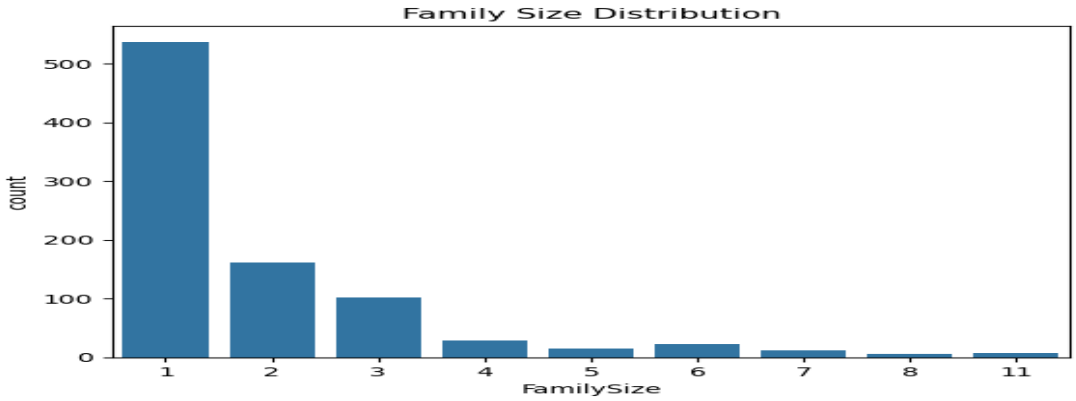
4.4 Age Distribution

The age distribution illustrates how passenger ages are spread across different ranges. The distribution indicates a higher concentration of young adults, with fewer children and senior passengers. This helps in later analyzing whether specific age groups received higher survival priority.



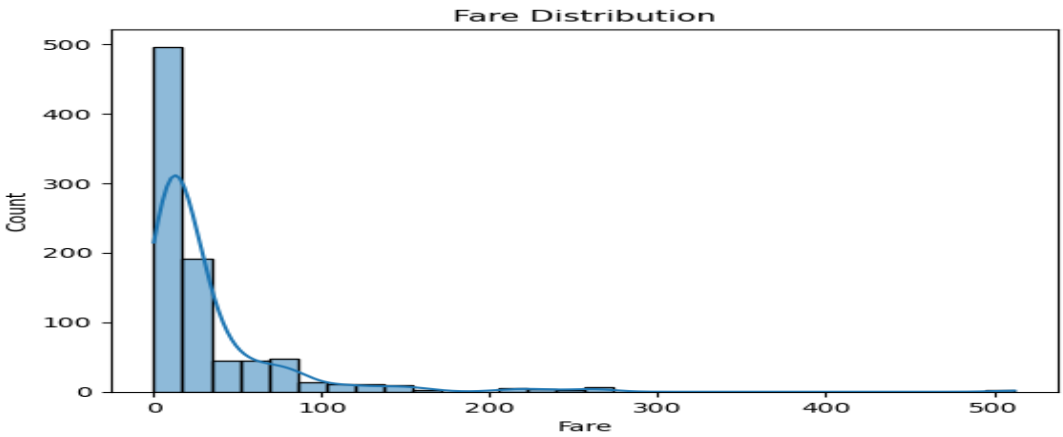
4.5 Family Size Distribution

The family size distribution shows how many passengers were traveling alone versus with family members. This feature helps evaluate whether traveling with larger or smaller families affected the ability to survive during the disaster.



4.6 Fare Distribution

The fare distribution is right-skewed, indicating that most passengers paid lower fares while a smaller group paid significantly higher fares. This suggests the presence of socio-economic disparity, which may indirectly relate to passenger class and survival likelihood.

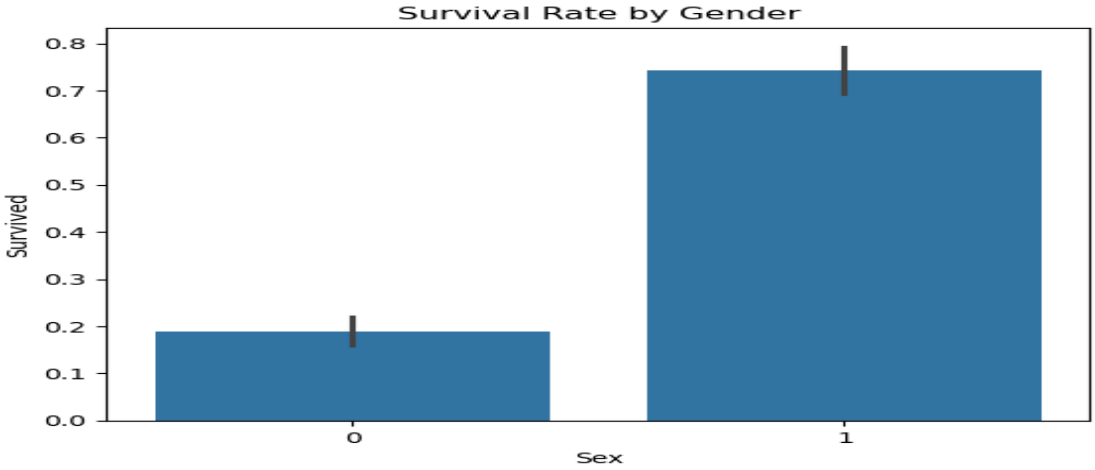


The univariate analysis provided an individual understanding of key features such as survival status, gender, passenger class, age, family size, and fare. These observations form the basis for deeper relational analysis in the next section, where interactions between variables and survival outcomes will be explored.

5. BIVARIATE ANALYSIS

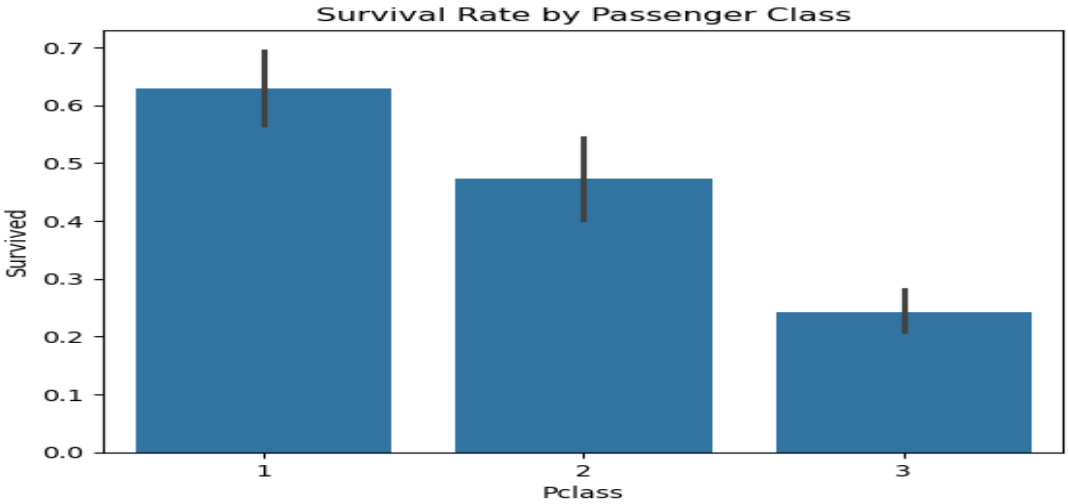
5.1 Survival vs Gender

The analysis shows that female passengers had a significantly higher survival rate compared to male passengers. This supports the evacuation priority rule of "women and children first," indicating that gender was one of the strongest factors influencing survival outcomes.



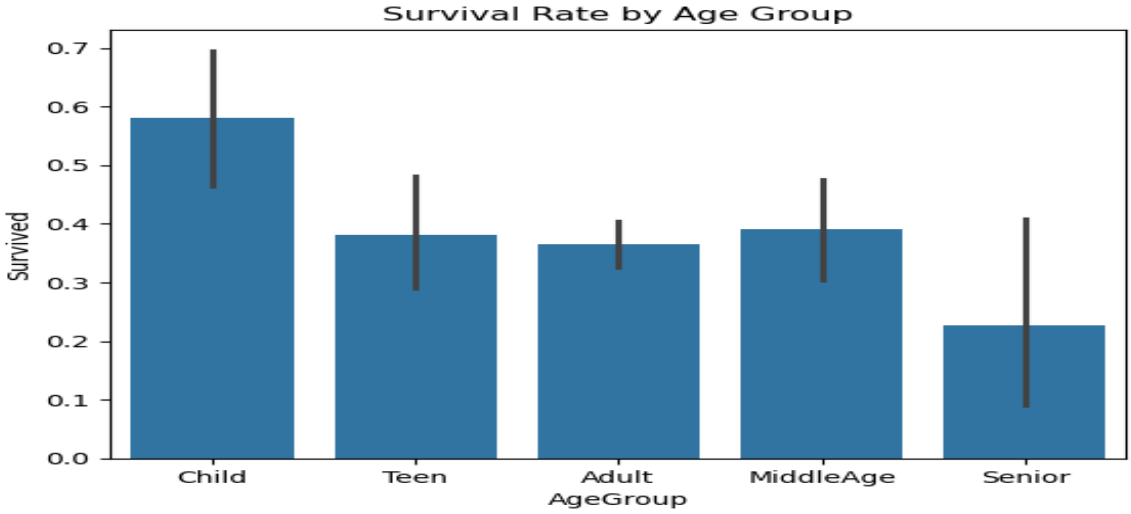
5.2 Survival vs Passenger Class

The survival rate is highest among first-class passengers and lowest among third-class passengers. This indicates that socio-economic status and cabin location played an important role in determining access to lifeboats and rescue opportunities.



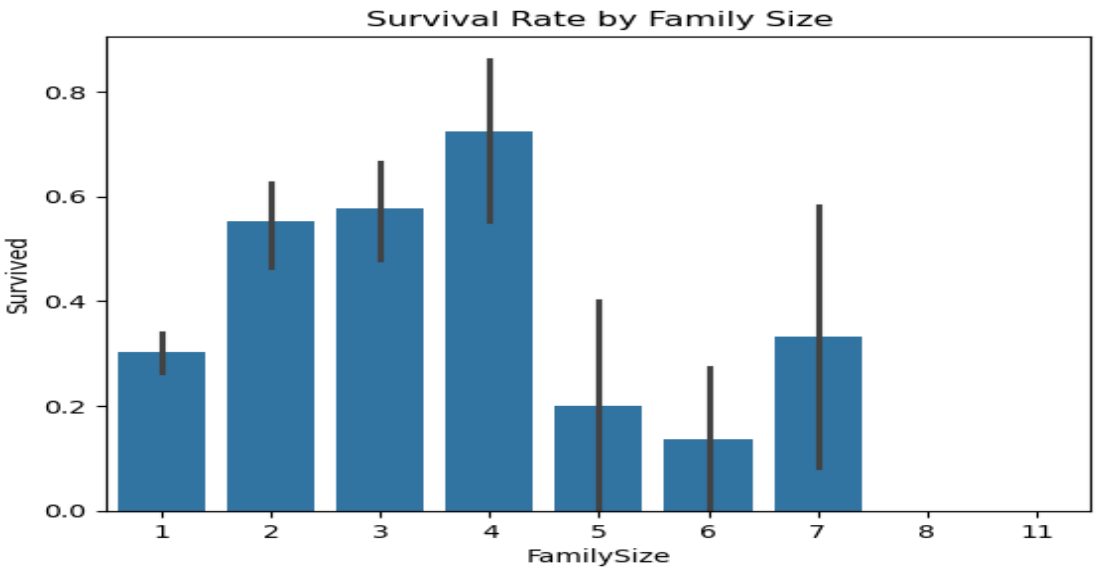
5.3 Survival vs Age Group

The analysis indicates that children had the highest survival chances, followed by adults, while senior passengers showed comparatively lower survival rates. This reflects evacuation priorities and physical mobility factors during the disaster.



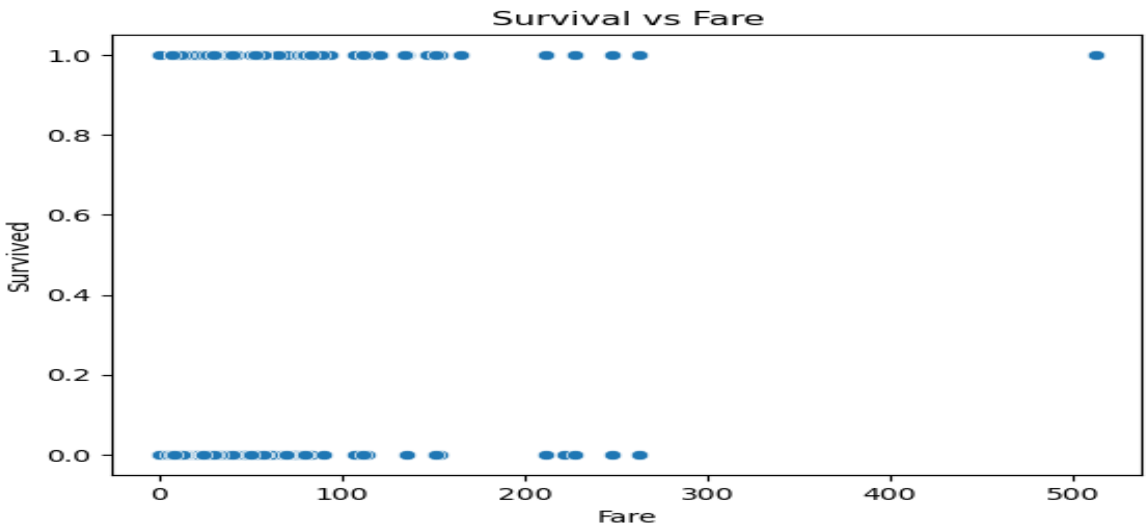
5.4 Survival vs Family Size

The relationship between family size and survival shows that medium-sized families had better survival rates compared to very small or very large families. Extremely large families may have faced coordination difficulties during evacuation, while passengers traveling alone lacked group support.



5.5 Survival vs Fare

Passengers who paid higher fares generally exhibited higher survival probabilities. Since fare is strongly associated with passenger class, this suggests that wealth and access to better cabin locations indirectly influenced survival chances.

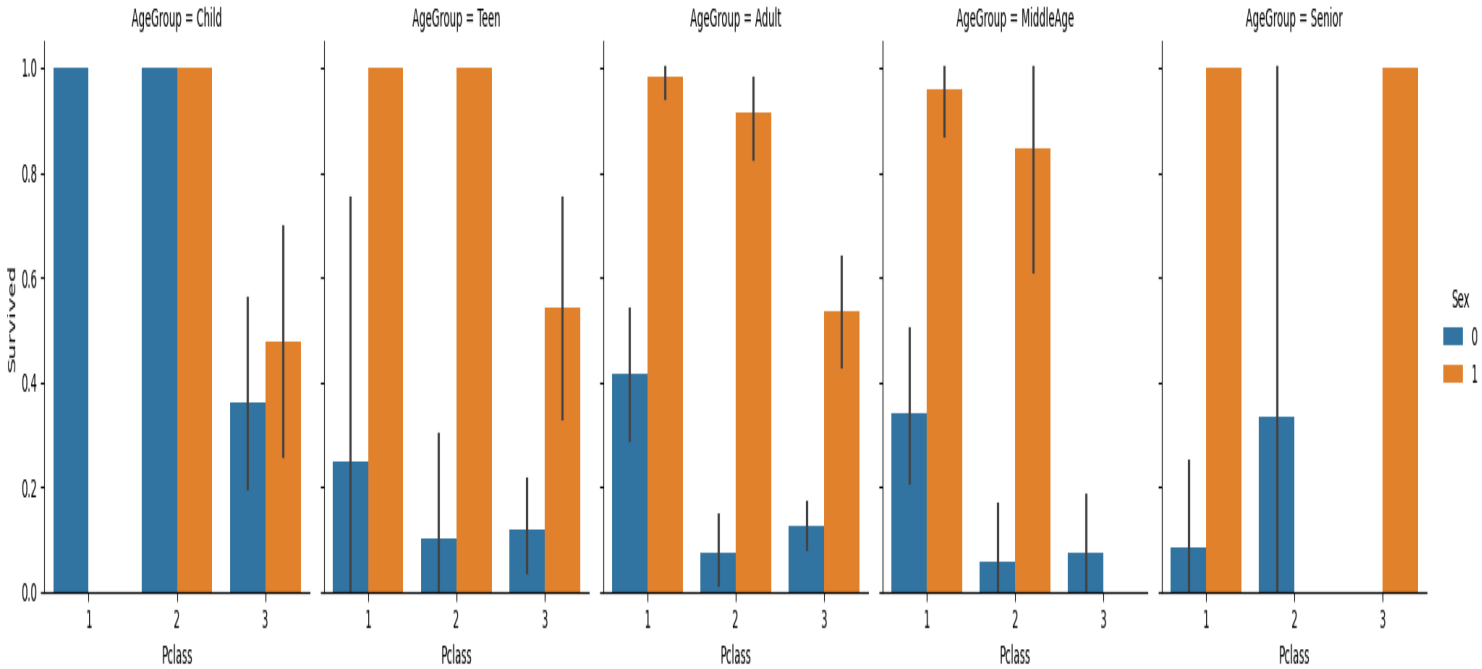


The bivariate analysis revealed that gender, passenger class, and age group were the most influential factors affecting survival. Family size showed a moderate influence, while fare indirectly reflected the effect of passenger class. These relationships motivated a deeper multivariate analysis to study how these variables interact simultaneously to influence survival outcomes.

6. MULTIVARIATE ANALYSIS

6.1 Combined Effect of Gender, Passenger Class and Age Group

The multivariate analysis reveals the combined influence of gender, passenger class, and age group on survival probability. The results show that female passengers consistently had higher survival rates across all classes and age groups. Among males, children had better survival chances compared to adult and senior males.



Passenger class further refined survival probabilities within each gender group. First-class females showed the highest survival rates, while third-class adult males exhibited the lowest survival chances. This demonstrates that survival was not determined by a single factor but by the interaction of multiple socio-demographic characteristics.

6.2 Hierarchical Survival Pattern

From the combined analysis, a hierarchical survival pattern can be observed. Gender appears to be the primary determinant of survival, followed by passenger class and then age group. Female passengers were prioritized for rescue, and within each gender category, first-class passengers had better access to lifeboats. Age group acted as a situational factor, where children were given priority over adults and seniors.

This hierarchy suggests that survival probability followed a layered decision structure:

1. Gender (Primary priority)
2. Passenger Class (Socio-economic access)
3. Age Group (Evacuation priority based on vulnerability)

Thus, survival outcomes were governed by a conditional interplay of multiple variables rather than a single dominant factor.

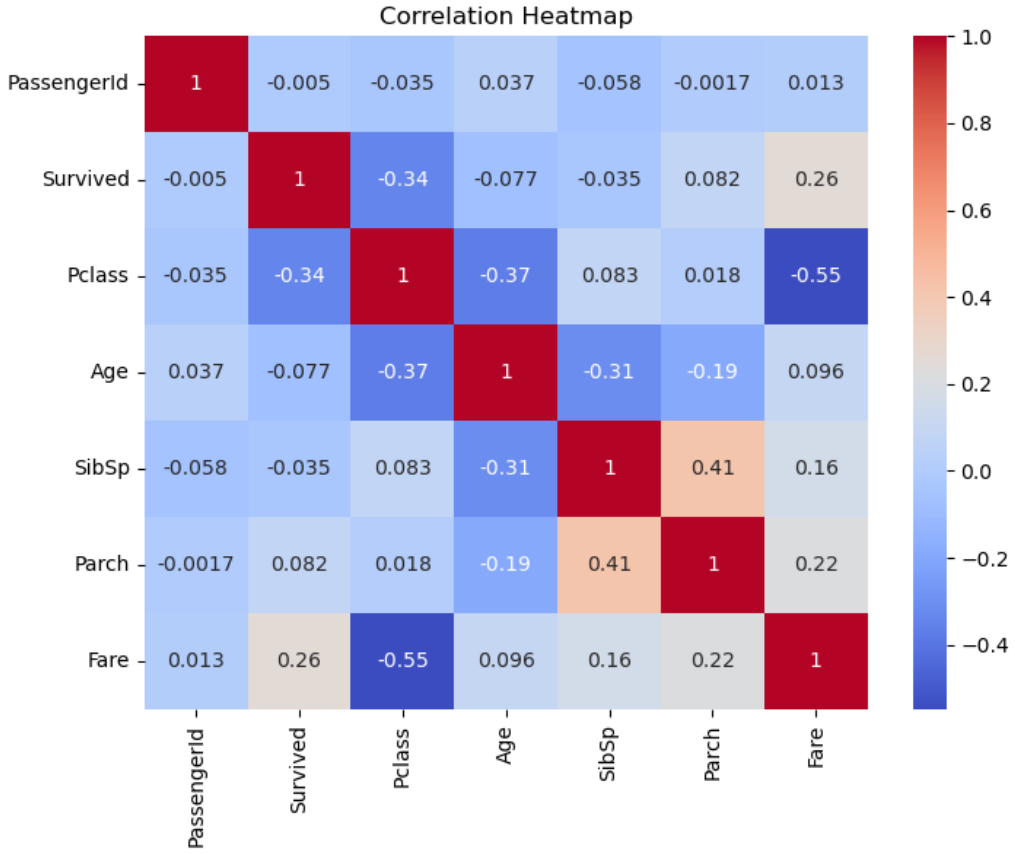
6.3 Exception Group Analysis

Certain passenger groups exhibited survival outcomes that deviated from general expectations, indicating the presence of conflicting factors. For example, adult male passengers in first class had better survival chances than adult males in third class, showing that socio-economic status could partially offset gender disadvantage.

Similarly, some female passengers in third class had lower survival rates compared to first-class females, indicating that access to lifeboats and physical proximity to rescue points influenced survival beyond gender priority alone. These exception groups highlight that survival rules were not absolute but dependent on situational and structural factors during evacuation.

6.4 Correlation Between Numerical Features

The correlation heatmap indicates that Fare has a positive correlation with survival, suggesting that higher-paying passengers had better survival chances. Passenger class shows a negative correlation with survival because higher class numbers represent lower socio-economic status. Age shows only a weak correlation with survival, implying that while age influenced priority decisions, it was not as strong a determinant as gender or class.



The multivariate analysis demonstrates that survival on the Titanic was influenced by a combination of gender, passenger class, and age group. Gender emerged as the most dominant factor, while passenger class refined survival access within gender groups. Age played a situational role, particularly benefiting children. The interaction of these variables reveals a hierarchical and conditional survival mechanism rather than a single-factor dependency.

7. KEY INSIGHTS AND CONCLUSION

- **Gender was the strongest predictor of survival.**
  Female passengers had a significantly higher survival rate compared to males, indicating a strong and policy-driven rescue priority.
- **Passenger class influenced survival chances.**
  First-class passengers showed the highest survival probability, followed by second class, while third-class passengers had the lowest survival rate.
- **Age showed a non-linear relationship with survival.**
  Children had relatively higher survival chances, whereas middle-aged adults formed the majority of casualties due to higher representation in the dataset.

- **Fare positively correlated with survival.**
  Passengers who paid higher fares (typically first class) had better survival outcomes, suggesting socioeconomic status played a role.
- **Port of embarkation had a noticeable impact.**
  Passengers embarking from Cherbourg showed higher survival rates compared to Southampton and Queenstown, possibly due to higher class distribution.
- **Family size affected survival probability.**
  Passengers traveling with small families (1–3 members) had better survival chances than those traveling alone or in very large groups.
- **Class, fare, and embarkation location were interrelated.**
  Higher fare and certain embarkation points were strongly associated with higher passenger classes, indirectly influencing survival.
- **Correlation analysis confirmed limited linear relationships.**
  Apart from fare and class, most variables showed weak linear correlation with survival, highlighting the need for multivariate analysis.
- **Missing values were concentrated in specific features.**
  Age and Cabin had significant missing data, which required careful preprocessing to avoid biased insights.
- **Overall survival pattern indicated social hierarchy influence.**
  Survival was not random; it was strongly associated with gender, passenger class, and economic standing.